College of computer and information science

My STC

Security Risk Management, Governance and Control (CYS403)

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Executive Summary

This report includes a security analysis and risk management plan of MySTC by STC Saudi Arabia, a large digital customer platform that is utilized to manage accounts, make bill payments, control packages, provide recharge services, follow-ups on complaints, and customer support. The platform is vulnerable to numerous cybersecurity risks, which can impact confidentiality, integrity, and availability due to the sensitive nature of the personal, service-related, and transaction-related information it is dealing with. The research analyzed the existing security situation of MySTC and discovered that it is conducted in an environment of technical controls, organizational actions, customer protection processes, and regulatory compliance requirements. Practical significance of solid online protection is indicated by the real and related evidence, which was given by the official STC reports, which proved the presence of recurring spam, fraudulent activities, and attempts of cyberattacks. The six information systems components were then used to develop a structured risk management plan which included hardware, software, network, data, people and procedures. It was determined that the most important aspects of the assessment are software, network, data, and people since failure in these aspects could directly impact customer confidence and platform security. Authentication and data confidentiality methods like OTP verification, account-based access control, organizational and technical measures of data protection were also tested in the report. Furthermore, it examined the security policy and customer guidelines of STC, such as privacy protection, fraud reporting, and complaint-handling processes. With the findings, a more effective security strategy was suggested to enhance the multi-factor authentication, enhance the protection of anti-phishing, secure mobile-based applications and APIs, broaden security monitoring, enhance incident response, and fortify governance and compliance practices. The report also indicated how these recommendations would enhance the CIA triad by ensuring protection of customer data, maintaining accuracy of the transactions and records and ensuring reliable service availability. On the whole, the analysis is that MySTC is a powerful and useful online platform, and ongoing security enhancement is needed to adequately meet the changing cyber threats. An effective security risk management strategy is hence necessary in a move to facilitate more trusted, safer, and resiliency digital telecom services in Saudi Arabia.

Introduction

The primary online customer portal of STC Saudi Arabia is MySTC, which is a significant mobile app, that allows users to control telecom services, gives the customer control over all the numbers registered under their name, pay bills, monitor usage, subscribe to packages, get customer support, and other account-related services, making it a high-value platform in terms of cybersecurity. Due to the sensitive nature of personal, account, billing and service data used in the application, the application encounters a major security challenge in regard to confidentiality, integrity and availability. Moreover, the privacy notice on STC states that the company handles various types of customer data and offers services with the help of MySTC and other digital platforms, making the enhanced significance of robust security rules and privacy measures. The site also facilitates the processing of customer complaints and follow-ups on services, indicating that the site is at the center of the online relationship between STC and consumers. Consequently, the vulnerability of authentication, access control, data protection, API security, or incident response might directly impact user confidence and service reliability. Given that telecom platforms are often the victims of fraud, phishing, identity abuse, and service outage, MySTC needs to uphold a robust and ever-advancing security stance. Thus, the given project is analysis of the present security situation of MySTC by STC, its main security problems, assessment of potential risks in all information system elements, and analysis of security policy and customer guidelines of the company. On the basis of that analysis, another idea that the report suggests is the implementation of a more secure strategy to minimize risks and improve the overall safety of the platform. Finally, this study aims to evaluate the potential of MySTC to enhance the security of user data and provide reliable and trusted digital services in Saudi Arabia.

Current Security Situation and Major Issues

Security Environment

MySTC today has both technical and organizational security environment that aims at securing customer accounts, personal data, billing activities, and operations of digital services. MySTC is a high-value telecom platform since it accepts bill payment, checking of balance, subscription of

packages, accessing support, and managing an account, and therefore, it involves sensitive information regarding users and transactions. The official documents of STC indicate that the platform is part of a more comprehensive privacy, authentication, customer-care, and regulatory framework, implying that its security is reliant on both technical controls and governance, legal compliance, and operational processes.

Examples of its **technical measures** include:

- **Authentication and Identity Verification:** STC provides the added security feature of one-time passwords to verify identity, confirm passwords, confirm payment, and protect financial transactions. This implies that verification controls that will minimize fraud and unauthorized activity support access to customer activities in MySTC.
- **Privacy and Data Protection Controls:** STC in its privacy notice indicates that it secures the personal data of customers both at the organizational and technical levels. The company further claims that they only store personal information as long as it is needed and subsequently destroys it securely in a way that meets the required laws and regulations.
- **Privacy and Security of User Data:** STC claims that it keeps the privacy and confidentiality of personal information and does not share it without a legal or authoritarian request. This indicates a technical and policy-driven strategy of safeguarding customer information processed via digital mediums like MySTC.
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Examples of its **organizational measures** include:

- **Internal Governance and Compliance**
- **Customer Support and Complaint Procedures**
- **Policies and Terms of Service**

Moreover, the attack surface of MySTC is very important as it gathers access to accounts, actions related to payment, processing of personal data, support, and service management activities

within a single digital platform. Consequently, it can be subjected to risks like phishing, account hacking, fraud, data breach, unauthorized access, and service interruption. This renders effective authentication, privacy management, handling of complaints and ongoing security checks particularly crucial in upholding customer trust and safe service provision.

Past and Recent Security Incidents

- **Fraudulent calls masquerading as STC representatives:** One of the previously reported security concerns about STC involved scammers posing as STC employees and calling customers to defraud them and inquire about sensitive information or money. STC had cautioned its customers against those unsolicited calls by scammers claiming that they were calling on behalf of the company. The significance of this kind of incident is that it demonstrates that the use of social engineering is a real and an active threat in the customer environment of STC over the course of many years and particularly where the trust in the brand of telecom can be used by an attacker.
- **2017 WannaCry-related security panic:** In the worldwide Wanna Cry ransomware pandemic in May 2017, STC declared that their networks and systems were not compromised but explained that what had been reported in the media was on a small number of personal devices that would be attended to by the technical teams. Although not a verified breach of the core systems of STC, still, it is a pertinent security incident since it demonstrates vulnerability to endpoint-level risk and the necessity of robust device security and patching, as well as expedited response mechanisms in a massive telecom setup.

Major Security Issues

Phishing, social engineering, and telecom fraud

Customer-focused fraud in the form of suspicious calls, SMS messages, and OTP abuse is one of the largest security challenges of MySTC and the broader digital landscape of STC. This is not merely a theoretical threat, but a practical and current one, as STC openly cautions clients against disclosing verification codes and has a special reporting form that allows reporting

suspicious and fraudulent messages or calls. This is a severe problem as malicious users may exploit the counterfeit STC messages to steal one-time passwords, credentials or payment-related data and proceed to carry out account theft or malicious transactions. Social engineering, phishing and telecom fraud.

High exposure to cyberattacks and ongoing attack pressure

According to Sustainability reporting of 2024, STC experienced 9,090, 9,708 and 8,405 attempted cyberattacks in 2022, 2023 and 2024 respectively and 26, 11 and 15 actual cyberattacks respectively. These numbers suggest that STC faces sustained pressure in the area of cybersecurity, and service disruption, unauthorized access attempts, and a digital platform compromise are some of the key security issues. Since MySTC is a fundamental customer-facing service, these attacks can impact directly on the availability of services, user trust, and continuity of operations.

Protection of sensitive personal and transactional data

MySTC processes a variety of sensitive data, such as registration data, payment data, customer support data, device identifiers, and certain location data. This poses a significant security concern regarding the confidentiality of customer information, since any failure in the access control, stored security, transmission security, or third-party services may put the users at risk of privacy breaches, frauds and identity abuses. The closer the data a platform handles is to the individual and the operational, the greater the need to ensure good data governance and robust technical protection.

Authentication and account takeover risk

Even though the OTP is an extra security feature applied by STC to verify the identity, payments and transactions, the fact that people are instructed not to share the OTPs publicly can only indicate that authentication is one of the primary risk areas. When users are tricked into providing verification codes by phishing, or by fraudulent support messages, attackers can obtain

unauthorized access to telecom accounts or sanction fraudulent activities. This renders the strength of authentication, awareness of users and secure process of transaction approval as a significant security concern to MySTC.

Compliance and cybersecurity governance risk

STC demonstrates high compliance with legal, regulatory and cybersecurity requirements, but not flawless compliance, with 99 in 2023 and 98 in 2024. Concurrently, it declares that its policy of cybersecurity addresses 19 security areas and is updated frequently to ensure confidentiality, integrity and availability. This implies that governance is a big focus, yet it is also a big security concern since a platform of this magnitude is supposed to be continually aligned between policies, legal requirements, technical constraints and day-to-day operations. The mere existence of small loopholes in governance may have big regulatory and operational impacts within a telecom environment.

Heavy dependence on digital channels and service availability

According to the reporting of STC itself, digital self-service support in STC KSA was 97% in 2024, indicating that the company heavily relies on digital channels to serve its customers and operate. This implies that MySTC and associated digital solutions are not auxiliary support systems, but rather, vital service channels. Consequently, outages, denial-of-service, app crashes, or significant disruption of the platform will become a serious security concern since they can instantly impact a very large client base and disrupt access to billing, customer service, and support.

Risk Management Plan

IS Component	Identified Risk	Why This Likelihood Was Chosen	Why This Impact Was Chosen	Likelihood	Impact	Risk Level	Control Strategy	Recommended Controls
Hardware	Server or infrastructure disruption caused by high traffic, service outages, hardware failure, or attack-driven overload affecting customer access to MySTC services.	The likelihood is **Medium** because hardware failure or overload may not happen daily, but MySTC depends heavily on digital service delivery and may face traffic spikes or attack-related pressure.	The impact is **High** because infrastructure disruption can prevent customers from accessing billing, recharge, support, and account services.	Medium	High	High	Preventive, Corrective, Physical/Infrastructure	Use redundant servers, backup infrastructure, disaster recovery planning, regular hardware maintenance, failover support, and high-availability architecture.
Software	Vulnerabilities in the MySTC mobile application, backend systems, APIs, or update mechanisms could allow unauthorized access, data leakage, fraud, or misuse of services.	The likelihood is **High** because mobile applications, APIs, and backend systems are highly exposed to attackers, especially when they support payments and account management.	The impact is **High** because software exploitation could expose customer data, allow account takeover, or affect transaction integrity.	High	High	Critical	Preventive, Detective, Technical	Apply secure software development lifecycle practices, regular vulnerability assessments, penetration testing, secure coding standards, patch management, API security testing, application monitoring, and code review.
Network	Network-based attacks such as denial-of-service, malicious traffic, spoofing, or unauthorized access attempts may affect service continuity and customer connectivity.	The likelihood is **High** because STC faces continuous cyberattack pressure, and telecom digital services are common targets for network attacks.	The impact is **High** because network disruption can affect MySTC availability and prevent customers from accessing essential digital telecom services.	High	High	Critical	Preventive, Detective, Technical, Corrective	Deploy firewalls, IDS/IPS, DDoS protection, network segmentation, traffic monitoring, real-time alerts, secure VPN access, and incident response procedures.
Data	Exposure, leakage, misuse, or unauthorized disclosure of customer personal data, payment-related information, device identifiers, support records, and service data	The likelihood is **High** because MySTC processes large amounts of sensitive customer and transaction-related information, making data a major target.	The impact is **High** because data leakage could cause privacy violations, fraud, identity misuse, regulatory penalties, and reputational damage.	High	High	Critical	Preventive, Detective, Administrative, Technical	Enforce encryption for data at rest and in transit, role-based access control, data minimization, secure retention and deletion, privacy audits, access logging, and compliance monitoring.

	processed through MySTC.							
People	Social engineering, phishing, fraudulent calls, fake messages, insider misuse, or customer disclosure of OTPs and credentials may result in unauthorized transactions or account compromise.	The likelihood is **High** because phishing, scam calls, OTP misuse, and social engineering are active threats in telecom environments and depend heavily on human behavior.	The impact is **High** because successful social engineering can lead to account takeover, fraud, unauthorized payments, and loss of customer trust.	High	High	Critical	Preventive, Detective, Administrative	Conduct employee cybersecurity training, customer awareness campaigns, OTP protection education, phishing simulations, least privilege access, suspicious activity alerts, fraud reporting channels, and insider access monitoring.
Procedures	Weak or inconsistent operational procedures in incident response, complaint handling, access approval, fraud escalation, or policy enforcement may delay detection and increase damage during security incidents.	The likelihood is **Medium** because procedures may exist, but gaps can appear when incidents are complex, urgent, or involve many departments.	The impact is **High** because weak procedures can delay response, increase service disruption, and reduce the effectiveness of technical controls.	Medium	High	High	Corrective, Administrative, Detective	Develop clear incident response procedures, fraud escalation workflows, access approval processes, complaint handling procedures, regular audits, staff accountability, communication plans, and post-incident review.

Control strategy

The main control strategy recommended for MySTC is risk mitigation. The table below summarizes the key risk areas, the appropriate control approach, and the controls needed to reduce each risk to an acceptable level.

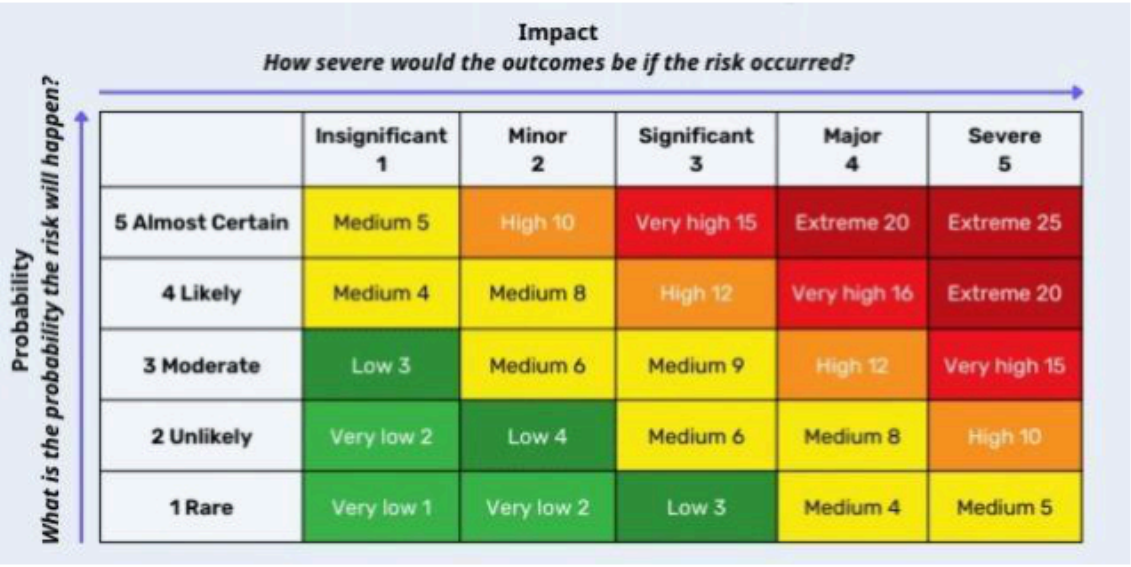
Risk Area	**Control Strategy**	**Recommended Controls**	**Expected Security Benefit**
Authentication and account takeover	Mitigation	Enforce stronger multi-factor authentication, protect OTP workflows, use adaptive login checks, and require additional verification for high-risk actions.	Reduces unauthorized access, OTP misuse, and account compromise.
Data privacy and confidentiality	Mitigation	Apply encryption for data at rest and in transit, role-based access control, secure retention and deletion, and regular privacy audits.	Protects customer personal, billing, and transaction data from leakage or unauthorized disclosure.

Phishing and telecom fraud	Mitigation	Provide customer awareness campaigns, in-app fraud alerts, easy reporting channels such as 330330, and monitoring for suspicious communications.	Lowers the success rate of social engineering attacks and improves fraud reporting and response.
Service availability and technical disruption	Mitigation	Use continuous monitoring, DDoS protection, backup systems, high-availability architecture, and tested incident response procedures.	Improves resilience and helps keep MySTC services available during attacks or failures.
Third-party and supplier risk	Mitigation	Perform vendor security assessments, enforce contractual security requirements, limit third-party access, and review compliance regularly.	Reduces exposure introduced by external service providers and strengthens overall governance.

Risk matrix

The risk matrix indicates that the most significant risks are connected with software, network, data, and people since they are associated with high probability and high impact. All these are areas that are directly related to customer accounts, payment, personal data, authentication, and service availability. Hardware and procedures are considered as high risks as they have medium chances, but high impact since they are crucial as they help to maintain reliable MySTC services. The suggested control strategies are therefore an integration of the preventive, detective, corrective, administrative, technical and physical / infrastructure controls to minimize the likelihood of incidents and duration of consequences should such incidents happen.

Risk matrix Table



Evaluation of Authentication and Data Confidentiality Techniques

One-Time Password (OTP) Verification

One-time password verification is one of the primary methods of authentication employed by STC in the world of the digital environment. According to STC, OTP is implemented as an extra security measure to verify identity of customers, confirm passwords, confirm payments and protect financial transactions. It is a powerful method since it incorporates an additional verification process prior to the completion of sensitive activities, thereby minimizing unauthorized transactions and misuse of customer accounts. Its efficiency, however, relies on the knowledge of customers greatly, since OTP-based security can be undermined with phishing, spamming support calls or fraudulent SMS messages which deceive a user to provide verification codes. All in all, OTP is a helpful and viable authentication control to MySTC, although it must be backed up by more effective anti-phishing tools and ongoing customer education.

Account-Based Access Control Through MySTC

The other significant authentication-related method is the controlled access model of the MySTC application where customers can directly view and administer all the numbers registered under their identity and view billing data, see usage, manage SIM-related data and take actions on accounts, all in a logged-in environment. This is a kind of account-based access control model whereby services are bound to the verified customer profile, and not freely available. The method is practical in that it allows the centralization of accounts administration in a single authenticated application and lessens the use of unauthenticated manual service requests. Concurrently, since MySTC allows accessing sensitive features, including online payments, saved cards, the records of bills, and SIM, the security of this model of access becomes highly significant. In case of a breach of login credentials or the verification of customer identity, much information about customers and functionality of their accounts can be revealed. Thus, it is an effective method of convenience in service, yet it demands high-quality session security, the

ability of secure login protection, and close attention to the presence of unauthorized access attempts.

Organizational and Technical Data Protection Measures

To ensure the confidentiality of data, STC publicly indicates that it applies organizational and technical data protection and data security procedures and measures to ensure that customer data does not get released or processed without authorization. The internal audit, external audit, staff and contractor privacy training and reporting to audit and data governance committees and regulatory authorities are specifically listed in the privacy notice. It further indicates that STC preserves the privacy and confidentiality of the personal data and only disclosed it when the consent or law allows it. This is an effective method of confidentiality since it integrates both procedural and operational controls as opposed to using a single control. Moreover, STC mentions fraud prevention and network and information security among the valid reasons to process some data, and this demonstrates that controlling confidentiality is interconnected with the more extensive security processes. The primary advantage of the method is that it aids in compliance, accountability and safeguarding of customer data throughout the service lifecycle. Its drawback is that the description aimed at the general audience is general and does not give a lot of technical information regarding implementation, and thus, as an outsider, the technique seems to be reasonable, but not completely transparent.

Analysis of Security Policies and Customer Guidelines

1. Privacy and Personal Data Protection Policy

The privacy policy of STC demonstrates a good formal dedication to the privacy of the customer information. According to the company, the company employs suitable organizational and technical security controls, such as internal audit, external audit, staff and contractor privacy training, reporting to audit and data governance committees and regulatory authorities. It also adds that, they will not divulge personal data without permission except in cases where disclosure is legally mandated, which translates to a policy framework that is focused on confidentiality, compliance, and controlled sharing of data. Moreover, the policy provides customers with such crucial rights as the right to be informed, the right to access their data, and

the right to alter their data via the Privacy section in MySTC, the right to request the erasure, and the right to withdraw consent. This is an encouraging move since it indicates that the customer guides of the company are not based on some vague statements, but are interconnected with certain user rights and privacy settings.

2. Customer Fraud Prevention and Security Awareness Guidelines

Customer guidelines of STC also speak of fraud and suspicious communications, which is extremely significant in a telecommunication setting where phishing, spoofed calls, and spam SMS messages constitute frequent threats. The company offers special way of letting customers manage unwanted promotional messages, as well as report suspicious, spam, or fraudulent SMS messages and calls by resending the number or message to 330330. This is a viable and practical customer-security guideline since it promotes active reporting as well as providing the customers with a direct avenue to take part in preventing frauds. Nevertheless, it also demonstrates that customer awareness is still a very crucial aspect of security, as the company will need to be dependent on users to detect and report on malicious messages instead of automatically eliminating all the abuse. All in all, the guideline is useful, but its success will greatly rely on the awareness and responsiveness of the customers in the face of the social engineering attempts.

3. Complaint Handling and Customer Rights Procedures

The complaint-handling policy of STC is a well-organized customer protection strategy that promotes accountability and trust. According to the company, complaints should be timely within 60 days of incident or disputed bill, complaints will be addressed within five days, and customers will receive a reference number, as well as written reply, either by text, electronic, or written communication. The policy further explains that the customers can raise and pursue their grievances via open platforms like MySTC, one-on-one chat, customer service desks, and the call center. This is a robust governance tool since it sets up a documented and traceable system of addressing customer grievances, even those that are customer service or security related. Security wise, this policy enhances transparency and aids in incident response though the actual success of the policy in reality relies on the rates of application and speed of these actions in practice.

Proposed Enhanced Security Approach

1. Stronger Multi-Factor Authentication

MySTC ought to enforce a more stringent multi-factor authentication during the process of log-in, approving payment, recovery of an account, and sensitive changes in the profile. This would minimize chances of account takeover, OTP abuse, and customer fraudulent activities.

2. Zero Trust Access Control

STC ought to implement Zero Trust to internal systems, administrative tools, and access by the third party. In this scheme, each user, device and session would be continuously authenticated prior to the issuance of access.

3. Advanced Anti-Phishing Protection

The site must enhance security against phishing, scam messages, counterfeit telephone calls and social engineering attacks on customers. Users would be able to evade security traps by using in-app warnings, fraud alerts, and simplified reporting options.

4. Secure Mobile App and API Development

MySTC needs to be reinforced with the use of secure coding, periodic penetration testing, API-level reviews, and expedited patching. This would minimize the possibility of attackers taking advantage of software vulnerabilities.

5. Continuous Security Monitoring

STC ought to increase real-time tracking of suspicious logins, fraud, and abnormal traffic, as well as service abuse. Around-the-clock oversight would enable the company to identify the threats at an earlier stage and react faster.

6. Stronger Data Encryption and Privacy Controls

Strict access control policies should support the strong encryption of customer data when storing and transmitting it, as well as when accessing it. This would enhance privacy of billing information, personal information and records of services.

7. Improved Incident Response Capability

STC ought to have a more robust incidence response procedure when it comes to fraud cases, service failures, cyber-attacks and data related incidents. A more coordinated and quicker reaction would lessen the damage and enhance service recovery.

8. Better Third-Party Security Management

This should be done by performing a regular assessment of any vendors, contractors or external service providers with MySTC. This would minimize the third-party vulnerabilities to the platform.

9. Expanded Security Awareness Programs

Phishing, OTP protection, scam calls, and safe online behaviors should be frequently educated to both the employees and the customers as a form of awareness. Enhanced awareness would mitigate human error, which is in most cases one of the weakest links in cybersecurity.

10. Regular Governance and Compliance Reviews

STC needs to regularly review its cybersecurity policies, privacy practices, and internal controls. The continuous governance and compliance audits would assist in maintaining security in accordance with the evolving threats and regulations.

How the Proposed Approach Improves the CIA Triad

Confidentiality

The suggested strategy enhances confidentiality by reinforcing the security of customer information, account information, billing information and transaction information stored or processed by MySTC. The presence of more strong multi-factor authentication, more efficient encryption, more strict control of access, anti-phishing would decrease unauthorized access and contribute to the situation when only confirmed users and authorized employees can access sensitive information.

Integrity

By ensuring the correctness and validity of customer accounts, payment activities, service records and online transactions, integrity would be enhanced. Secure coding, API security, constant monitoring, and Zero Trust access control would assist to prevent inappropriate data modification, fraud, and misuse, and audit and governance reviews would contribute to trust in the appropriateness of system functions.

Availability

The suggested solution would enhance its availability by enabling MySTC to be available and reliable even in cases of cyberattack, fraud or technical disturbances. Ongoing observation, enhanced response of the incidents, enhanced third-party supervision, and enhanced business continuity support would minimize downtime and help in providing stable digital services to the customers.

Impact on the CIA Triad

On the whole, the suggested improved security strategy establishes a more balanced model of protection to MySTC since it encompasses the three key pillars of cybersecurity. It helps avoid the exposure of customer information, maintain the accuracy of data and transactions and assists in ensuring that telecom services are available when the customer requires them.

Conclusion

MySTC by STC is an extremely significant digital customer platform in Saudi Arabia as it facilitates the necessary services, including account management, bill payment, recharge, package management, customer support, and complaint management, all via one application.. The sensitive nature of such services means that the platform handles sensitive personal, transactional, and service-related data and thereby, cybersecurity is a significant operational need. Analysis revealed that MySTC operates in a security environment that encompasses technical protection, privacy, organizational, customer protection, and regulatory requirements. Simultaneously, the actual data provided by official STC reports and other sources proved that the platform is under constant pressure due to spam, fraudulent activities, phishing, and cyberattacks. Such threats demonstrate that such threats to MySTC are not only theoretical but

real and current in terms of their practical aspect. The significant security risks that were found in the report were fraud and social engineering, exposure to cyberattacks, personal data protection, authentication vulnerabilities, governance risk, and service availability. These risks had to be categorized and assessed in the six information systems components and as such a structured risk management plan was required. The proposed improved security system tackled these problems by implementing better authentication, Zero Trust access control, and enhanced anti-phishing, secure software development, active monitoring, enhanced incident response, enhanced privacy, and enhanced governance practices. This would enhance confidentiality to protect the information of customers, enhance integrity by maintaining correct transactions and account records and enhance availability by reducing disruption and downtime. The general conclusions are that MySTC already has a structured security and compliance framework, although the increasing size of digital services implies that more robust and flexible cybersecurity is still necessary. With the trust that customers place on digital telecom platforms growing, staying trustworthy will rely on how fast and well STC can handle the emerging threats. This is why constant monitoring, periodic review of policies, greater technical controls, and awareness of users should be among the priorities. Finally, a holistic security risk management plan will ensure that MySTC is safe, assuring and resilient to the emerging cyber threats.

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